

## Practical 6

### Rendering External JSON Data using Fetch API, Search and Filter

1. How is JSON fetched, parsed, and rendered?

Ans: JSON is fetched using the Fetch API. First, JavaScript uses `fetch()` to get the JSON file. Then the response is converted into JavaScript data using `.json()`. After parsing, the data is used to create HTML elements and display the information on the webpage.

So, the basic process is:

Fetch JSON → Parse JSON → Use Data → Render on Webpage

2. Which array methods are used for search, filter, sort, and pagination?

Ans: For searching, filtering, sorting, and pagination, different JavaScript array methods are used.

Search: `find()` or `findIndex()`

Filter: `filter()`

Sort: `sort()`

Pagination: `slice()`

These methods help to easily manage and display JSON data on the webpage.

3. How are loading and error states handled?

Ans: A loading message is shown while fetching data. If fetching is successful, the data is displayed. If an error occurs, an error message is shown using `catch()`.

#### 4. How is modularity maintained in JavaScript files?

Ans: Modularity is maintained by dividing JavaScript code into separate files and functions, where each file handles a specific task.

#### Screenshot:

The screenshot displays the 'StudentHub - Events' page. At the top is a dark blue header with the title 'StudentHub - Events' in white. Below the header is a light gray navigation bar containing links: Home, About, Register, Login, Dashboard, Events, Profile, Contact, Admin, and FAQ. The main content area is white and features a section titled 'Events List'. This section includes a search bar with the placeholder 'Search...', a dropdown menu for 'All Venues', and a dropdown menu for 'Sort by Date'. Below these filters is a table with three columns: 'Date', 'Event', and 'Venue'. The table contains three rows of event data. At the bottom of the table, there are 'Prev' and 'Next' buttons flanking the text 'Page 1 of 1'. The footer of the page is a dark blue bar with the copyright notice '© 2026 StudentHub' in white.

| Date        | Event                    | Venue          |
|-------------|--------------------------|----------------|
| 10 Aug 2026 | Web Development Workshop | Seminar Hall   |
| 18 Aug 2026 | Hackathon                | Computer Lab   |
| 25 Aug 2026 | Sports Day               | College Ground |

## StudentHub - FAQ

HomeAboutRegisterLoginDashboardEventsProfileContactAdminFAQ

### Frequently Asked Questions

All Categories ▼Default Order ▼

**Q: What is StudentHub?**

A: StudentHub is an online portal for managing attendance, assignments, results, notices, and student profiles.

Category: General

**Q: How do I register?**

A: Go to the Register page, fill in your details, and submit the registration form.

Category: Account

**Q: How do I log in?**

A: Open the Login page and enter your registered email and password.

Category: Account

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### Frequently Asked Questions

#### Q: Where can I check my results?

A: You can view your results from the Dashboard after logging in.

Category: Academics

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## StudentHub - Profiles

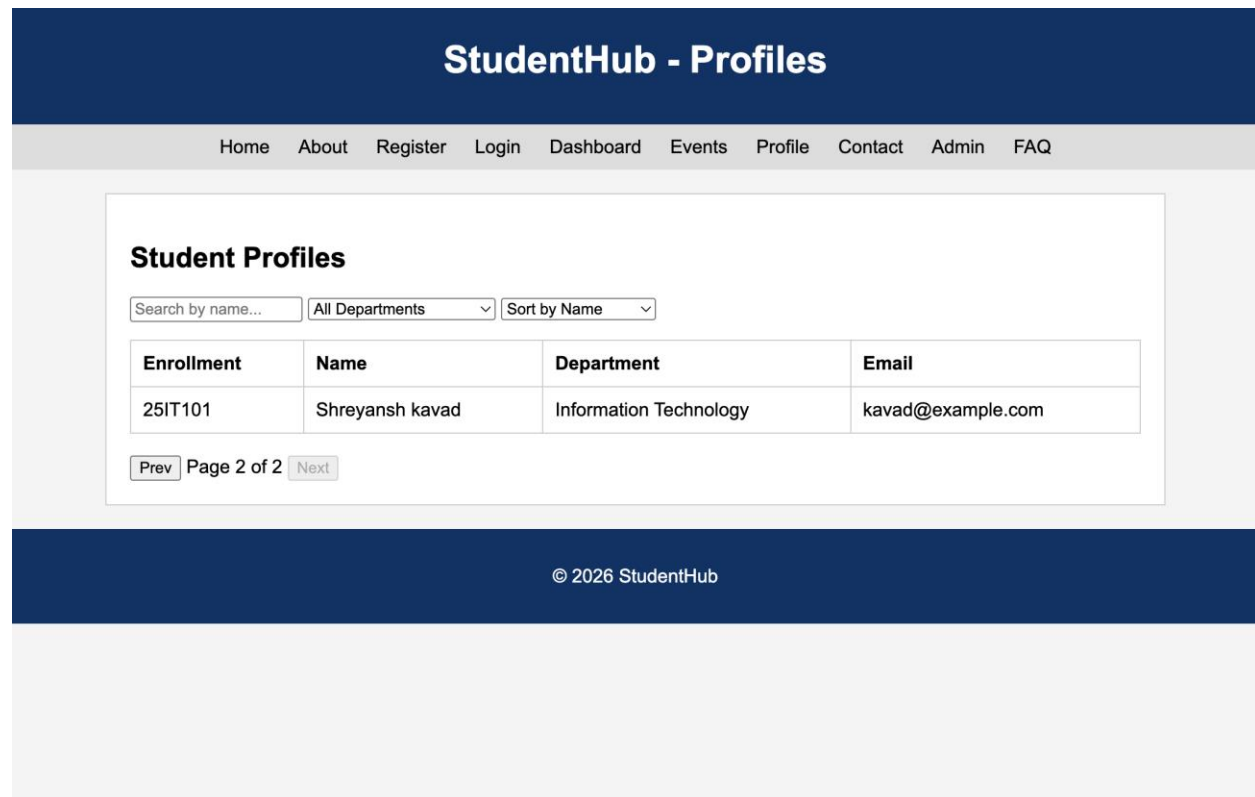
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### Student Profiles

| Enrollment | Name           | Department           | Email                   |
|------------|----------------|----------------------|-------------------------|
| 25CE101    | Jay patel      | Computer Engineering | jay@example.com         |
| 25DCE055   | Rushant Maniya | Computer Engineering | rushant2528@example.com |

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### GitHub Link:

<https://github.com/rushantmaniya/WDF/tree/f71155080218de5230827827f41a6842ecfad263/Practical-6>

### Challenges Faced:

I faced some difficulty in fetching and reading the JSON data correctly. It was difficult to apply search, filter, and sorting together. Like how to update HTML code and how to apply search operation and fetching. I faced problems while implementing pagination for the data.

### Learning Outcome:

I learned how to fetch and display JSON data using Fetch API. I also learned search, filter, sorting, and pagination using JavaScript.